

High-precision brain tumor classification from MRI images using an advanced hybrid deep learning method with minimal radiation exposure

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ARTICLE INFO

Keywords:

Hybrid modelling
Radiation risk reduction
Deep learning
Brain tumor
MRI images

ABSTRACT

Background: Accurate identification of brain tumors is critical to improving patient outcomes and minimizing unnecessary radiation exposure from imaging procedures. While Magnetic Resonance Imaging (MRI) is the gold standard for brain tumor detection, manual interpretation remains time-consuming, error-prone, and subject to inter-observer variability.

Objective: This study aims to develop a high-precision, automated MRI-based brain tumor classification model using a hybrid deep learning architecture to reduce diagnostic errors and support radiation exposure minimization strategies.

Methods: A novel hybrid deep learning model was developed by integrating the CE-EEN-B0 and ResGANet architectures. The model incorporates advanced feature selection and ensemble-based learning techniques to enhance classification performance across diverse datasets. The feature vectors extracted from MRI images were benchmarked against state-of-the-art (SOTA) deep learning classifiers, including InceptionV3, Vision Transformer, MobileNet, VGG-SCNet, DenseNet121, and ResNet50.

Results: The proposed hybrid model achieved an accuracy of 99.11 %, with a precision, recall, and F1-Score of 99.6 %. It also attained a specificity of 99.75 %, an error rate of just 0.01, and a Cohen's Kappa score of 99.10, outperforming all benchmark models.

Conclusion: The hybrid CE-EEN-B0-ResGANet model demonstrates high reliability and performance in MRI-based brain tumor classification. Its strong diagnostic metrics support its potential for clinical deployment as an effective, automated tool for aiding radiologists and minimizing unnecessary imaging interventions.

1. Introduction

The human brain is a significant part of the nervous system and works in conjunction with the spinal cord as the central part of the cavity of the neural tube to form the central nervous system (CNS) (Qureshi et al., 2022; Vaccarino et al., 2025). The brain acts as a control centre for many vital functions in the body, including analysis, integration, decision-making, and command transmission to other organs. The intricate anatomical structure of the human brain makes diagnosis of

neurological disorders particularly complex (Zahoor et al., 2022). Common CNS conditions such as strokes, infections, brain tumors, and chronic headaches pose considerable challenges in diagnosis, treatment planning, and prognosis (Arabahmadi et al., 2022). A brain tumor is defined as an abnormal growth of cells within the skull that can exert pressure and cause functional impairments (Rasheed et al., 2023a; Tandel et al., 2019). Even a small expansion within the limited intracranial space can lead to severe consequences (Borole et al., 2015). Brain tumors are among the top ten causes of cancer-related mortality,

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<https://doi.org/10.1016/j.jrras.2025.101858>

Received 3 July 2025; Received in revised form 30 July 2025; Accepted 4 August 2025

Available online 13 August 2025

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affecting both adults and children (Iorgulescu et al., 2023). Their impact on survival rates depends on a range of factors including type, size, location, and malignancy grade (Tiwari et al., 2022).

Each year, about 250,000 individuals are diagnosed with brain tumors, with 2 % being identified as malignant (Tiwari et al., 2022). In 2020, an estimated 23,890 adults in the United States were expected to be diagnosed with brain tumors 13,590 males and 10,300 females. Similarly, Australia reported 1879 new cases in the same year. Primary brain tumors affect about 14.1 % of Americans annually, with 70 % of these occurring in children. Despite significant clinical attention, early detection and effective treatment remain limited (Anaya-et al., 2022). From 2004 to 2020, brain tumor incidence increased globally from approximately 10 %–15 % (Xie et al., 2022). There are around 130 known types of brain tumors, broadly classified into primary (originating in the brain) and secondary (resulting from metastasis). Primary tumors are further divided into benign and malignant categories (Mehekare et al., 2017; Rasheed et al., 2024a). whereas secondary tumors are always malignant and often life-threatening (Rasheed et al., 2024b; Soomro et al., 2022).

Magnetic Resonance Imaging (MRI) is a widely used non-invasive imaging technique renowned for its ability to produce high-resolution images of soft tissues, making it particularly effective in detecting and evaluating brain tumors (Abu Mhanna, 2024). By using strong magnetic fields and radio waves, MRI can differentiate between healthy and abnormal brain tissues, offering detailed anatomical visualization that is essential for initial diagnosis, tumor localization, and monitoring treatment response (Ahmad et al., 2021). Its capability to provide contrast between various types of brain matter without exposing patients to ionizing radiation makes it the preferred imaging modality in neuro-oncology (Ahmad et al., 2020a, 2020b). MRI is also crucial in identifying tumor margins, edema, hemorrhage, and necrosis, which are vital for accurate tumor grading and surgical planning.

Functional Magnetic Resonance Imaging (fMRI), on the other hand, provides complementary information by measuring changes in blood oxygenation and flow that occur in response to neural activity (Abu Mhanna, 2024). This allows clinicians to map functional areas of the brain such as those responsible for motor control, speech, and sensory processing prior to surgical intervention. fMRI is especially valuable in planning surgeries near eloquent brain regions, as it helps minimize the risk of postoperative neurological deficits (Alnawafleh et al., 2024). Together, MRI and fMRI form a comprehensive diagnostic and planning toolkit that supports precise, patient-specific treatment strategies. Their integration into clinical workflows enhances the ability of medical teams to make informed decisions, improve surgical outcomes, and preserve critical brain functions during tumor resections (Mhanna et al., 2025; Oglat, 2022).

According to a recent study, brain tumors make up around 85–90 % of all primary CNS cancers. Accurate early detection is critical to improving patient outcomes and reducing mortality from brain tumors (Zhang et al., 2016). Advances in medical imaging, particularly MRI, have significantly enhanced diagnostic capabilities (Amin et al., 2025; Mamun et al., 2022a; Wang et al., 2025, p. 451). However, manual analysis of MRI scans by radiologists remains time-consuming, expertise-dependent, and prone to errors (Chatterjee, 2021; Li et al., 2025). Automated, computer-aided diagnosis using Artificial Intelligence (AI) and Deep Learning (DL) offers a promising solution to these limitations (Athamnah et al., 2024; Rasheed et al., 2023b). Recent technological advancements have significantly influenced the healthcare industry, particularly in AI and machine learning (ML) (Mamun et al., 2022b). DL is frequently utilized in healthcare for analysis, classification, and detection (Fayaz et al., 2025; Usman et al., 2017). Convolutional Neural Networks (CNNs), inspired by the human brain’s ability to recognize spatial patterns, are widely adopted in medical image processing. Well-known CNN architectures include ResNet, GoogLeNet, AlexNet, and VGG (Lin et al., 2015). Its computational capability is modeled after the human brain. Just as humans recognize objects by appearance, CNNs

are notably effective at processing images. The CNN models that stand out are ResNet with 152 layers, GoogLeNet with 22 layers, AlexNet with 8 layers, and VGG with 16–19 layers (Musallam et al., 2022). Table 1 provides a comparative overview of state-of-the-art (SOTA) ML and DL techniques applied to MRI and computed tomography (CT) images for brain tumor classification.

The central hypothesis of this study is that combining Contour Extraction Based Extended EfficientNet-B0 (CE-EEN-B0) and Residual Group Attention Network (ResGANet) within a hybrid framework augmented with a genetic programming (GP)-based denoising method and dynamic ensemble decision-making will significantly outperform traditional DL models in terms of classification accuracy, recall, specificity, and other evaluation metrics.

This study contributes a novel approach that distinguishes itself from previous work in several ways. Unlike conventional models that rely on single CNN architectures, this research introduces a hybrid ensemble model combining the feature-rich capabilities of CE-EEN-B0 and the attention-enhancing depth of ResGANet. Furthermore, the model incorporates a GP-based filter to effectively remove Rician noise, improving input data quality before classification. Finally, the study proposes a dynamic hybrid DL algorithm with regularization and classifier diversity mechanisms offering a new direction for medical image classification models. To the best of our knowledge, this is the first time such an integrated approach has been used for brain tumor classification using MRI images.

This article aims to improve the accuracy of brain tumor detection using the proposed Hybrid DL model integrating CE-EEN-B0 and ResGANet. The study focuses on reducing classification errors and improving the generalisability of the model through ensemble learning and advanced feature extraction. The proposed method is rigorously evaluated against leading DL models including ResNet50, DenseNet121, MobileNet, Vision Transformer, VGG-SCNet, and InceptionV3. The following are the significant contributions of this study:

- This study suggested an innovative and efficient hybrid DL model integrating advanced CNN architectures CE-EEN-B0 and ResGANet to classify glioma, meningioma, pituitary, and no tumor using MRI images. In addition, the proposed method deploys an incredible fusion descriptor to extract momentous features using MRI images.
- The proposed methodology for MRI image analysis was evaluated using various DL models, including ResNet50, DenseNet121,

Table 1
The comparison of SOTA ML and DL approaches for MRI and CT brain image analysis.

Previous Work	Dataset	Methods	Accuracy (%)
A.S. Musallam et al. (Musallam et al., 2022)	3394 MR Images	DCNN	97.072
M. Woźniak et al. (Woźniak et al., 2023)	3064, CT Brain Scan Images	CLM + CNN	96
J. Amin et al. (Amin et al., 2022)	2556 Brain Tumor Images	KNN-RF-DT	97.30
D.R. Nayak et al. (Nayak et al., 2022)	3260 MRI Brain Images	Dense Efficient Net	98.78
M.R. Obeidaviet al. (Obeidavi et al., 2022)	BRATS 2015 Dataset, 2000 MR Images	Residual Network	97.05
H.A. Halil et al. (Khalil et al., 2020)	BRATS 2017, 3D Image Dataset	Modified Two-Step Dragonfly Method	98.20
Proposed Work	6420 MRI Tumor Dataset	Hybrid System Model integrating CE-EEN-B0 and ResGANet architectures	99.11

MobileNet, Vision Transformer, VGG-SCNet, and InceptionV3. The analysis used Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG) descriptors. With a fusion feature descriptor, the proposed system model outperformed the existing DL algorithm in accuracy. Furthermore, the genetic programming model is deployed to denoise Rician noise, outperforming other approaches such as non-local means (NLM) and the fusion method.

- The model significantly reduces false positive and negative results, enhancing reliability by utilizing ensemble methodologies and careful feature selection, thereby tackling a crucial hurdle in medical diagnostics.
- The experiments involve the utilization of MRI scans of Glioma, Pituitary Meningioma, and No Tumor. Results show that the proposed system model outperforms other methods in accuracy, specificity, precision, F1-Score, recall, Cohen's Kappa, and error rate.

2. Literature review

The application of DL and ML in medical imaging, particularly for brain tumor detection, has significantly evolved over the past decade. Brain tumors are complex in their structure and variability, making early and accurate diagnosis critical for improving patient outcomes. Among the various imaging modalities, MRI remains the most preferred due to its superior soft tissue contrast and non-invasive nature. However, manual analysis of MRI scans by radiologists, although standard practice, is subject to variability and human error. This has led to the development of automated systems based on AI and DL models to assist in clinical decision-making.

2.1. Recent developments in deep learning for brain tumor detection

CNNs have emerged as the foundational architecture in DL for image-based classification tasks. CNNs are particularly effective in image analysis due to their ability to automatically extract features from raw data, which is crucial in medical imaging where data is highly complex (Rayed et al., 2024). Pioneering models such as AlexNet, VGGNet, ResNet, and GoogLeNet have demonstrated remarkable performance in general image recognition and have been successfully adapted for brain tumor classification (Nahiduzzaman et al., 2025). For example, Musallam et al. (Musallam et al., 2022) applied a deep CNN to a dataset of over 3000 MRI images, achieving an accuracy of 97.72 %. Similarly, Nayak et al. (Nayak et al., 2022) utilized a Dense EfficientNet model for MRI classification, reporting an accuracy of 98.78 %. These models have significantly contributed to improving diagnostic accuracy and reducing the time required for tumor detection.

2.2. Hybrid and ensemble learning approaches in medical imaging

Despite the achievements in the application of CNNs, there are issues that still need solving. One of the challenges lies with medical datasets, which are characterized by class imbalance. For example, in the Medical Imaging Datasets the commonly occurring class of benign tumors overshadows the infrequent class of malignant tumors. This imbalance may cause bias in the model leading to suboptimal performance in other datasets. Also, MRI scans are often susceptible to noisy artifacts, and Rician noise in particular, which diminishes the quality of the images and the model's accuracy. To overcome the limitations associated with the traditional CNNs, separate architecture and techniques are used in a hybrid model which aims to improve the accuracy of classification.

Hybrid models capitalize on the advantages of various DL architectures to boost classification accuracy. For example, Rasheed et al. (Rasheed et al., 2024a) combined CNNs with attention mechanisms, enabling the model to better focus on important features, such as the tumor area in MRI images. Other studies have examined ensemble approaches, where predictions from multiple models are merged to enhance overall performance. Additionally, preprocessing methods like

NLM filtering and wavelet-based noise reduction have been applied to improve image quality before being processed by DL models (Kalaiselvi et al., 2022). These techniques help address noise-related challenges and increase the model's resilience. However, despite the individual success of hybrid models, noise reduction techniques, and ensemble strategies, there remains limited integration of these approaches into a unified framework in the current literature (Amin et al., 2024).

2.3. Challenges in brain tumor classification and image processing

Despite the advances in DL, several challenges persist in brain tumor classification (Alam et al., 2025). Class imbalance, inadequate generalization across different datasets, and the presence of noise in images are major challenges that undermine the effectiveness of current systems. Although some models have utilized techniques like data augmentation and transfer learning to address issues with dataset limitations, and others have implemented attention mechanisms to enhance feature learning, these models still face limitations in terms of overall accuracy and reliability in practical settings. Furthermore, the scarcity of annotated data makes it even more difficult to train models that are both accurate and robust.

2.4. Motivation for the proposed hybrid model

This research presents a novel hybrid deep learning approach that combines the CE-EEN-B0 and ResGANet architectures, specifically designed for efficient feature extraction and tumor classification. The synergy of CE-EEN-B0's robust feature extraction and ResGANet's attention mechanisms enables the model to achieve superior accuracy and reliability in detecting brain tumors. Additionally, a GP based filter is applied to remove Rician noise, significantly enhancing the clarity of MRI images before classification. To further improve performance across different tumor types, a dynamic ensemble decision-making strategy is employed, increasing the accuracy of classification across various categories of brain tumors.

In contrast to previous methods, this work uniquely integrates SOTA DL techniques, advanced preprocessing methods, and ensemble learning into a single cohesive model. This combination overcomes the limitations of earlier models by improving noise robustness, reducing classification errors, and enhancing performance across multiple evaluation metrics. Consequently, the proposed model offers the potential to significantly advance the accuracy and consistency of brain tumor classification from MRI scans.

3. Materials and methods

3.1. Proposed architecture

In this research, the Hybrid DL Model was utilized with two advanced CNN architectures, CE-EEN-B0 and ResGANet, both pre-trained on the ImageNet dataset. These models were chosen for their strong feature extraction capabilities. A multi-stage CNN approach inspired by MTCNN (Multi-task Cascaded Convolutional Networks), a DL-based framework initially designed for object detection and alignment, was also employed. MTCNN operates in three stages (P-Net, R-Net, O-Net) to propose, refine, and output bounding boxes and landmarks. This hierarchical detection strategy was adapted for brain tumor detection to propose, refine, and detect tumor regions in medical images. CE-EEN-B0 is renowned for its exceptional processing efficiency and remarkable accuracy, processing input images of dimensions $224 \times 224 \times 3$ through convolutional layers, Swish activation layers, and batch normalization, as shown in Fig. 1.

The network adaptively modifies the layers' depth, width, and resolution in a balanced way. CE-EEN-B0 successfully captures many features from the input image by employing this method. It adjusts its complexity based on the existing resources and task requirements.

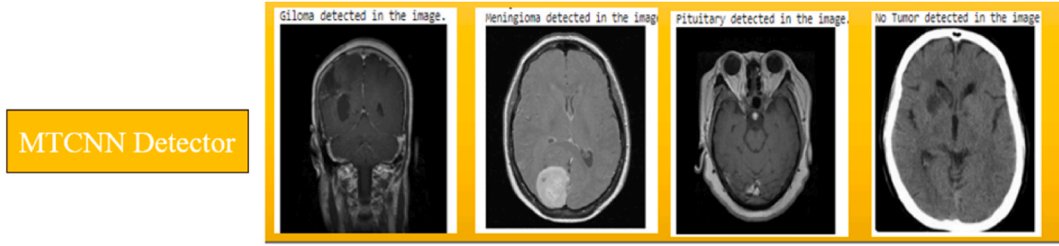


Fig. 1. Detection of brain tumors using MTCNN detector.

ResGANet features skip connections, also called residual connections, which allow gradients to be directly propagated throughout the network's layers. This model utilizes convolutional layers, ReLU activation, and batch normalization to process the input. The vanishing gradients problem is effectively addressed using skip connections, enabling the training of considerably deeper networks.

Both approaches possess initial layers that capture essential components like edges and textures. As the layers proceed, they extract more complex patterns crucial for specific tasks, such as identifying or classifying objects. The output produced by these networks is then passed through further specialized layers, including a flattening layer that converts the 2D feature maps into a 1D vector. Subsequently, there are dense layers that perform extra steps and diminish the features. The outcome is set at a softmax layer, which produces a probability distribution over the target classes. The ensemble model is created by combining the capabilities of CE-EEN-B0 and ResGANet. The model undergoes training, validation, and testing for 50 epochs. The training is conducted with a batch size of 32 and a learning rate 0.001. The model employs the Adam optimizer and a categorical cross-entropy loss function to optimize for accurate classification across several categories.

Our research methodology encompassed several vital phases. Initially, the data was obtained from an internet source accessible on [Kaggle.com](https://www.kaggle.com). After collecting the data, a GP model is presented for reducing Rician noise to pre-process the dataset. To effectively mitigate Rician noise, this GP approach combines a Feature Extraction component, GP Optimal Expression, and an Optimum Removal Estimation component. Moreover, data augmentation is used with diverse and

diversity of a dataset by applying various transformations to the images. This helps improve the model's generalization and robustness by exposing it to a broader range of variations in the input data. Rotating the image by a certain angle allows the model to recognize objects from different orientations. Flipping the image horizontally or vertically helps the model learn to generalize better to variations in object orientation. Scaling: Resizing the image to a different size helps the model learn to recognize objects at various scales. Brightness adjusting the brightness of the image allows the model to learn to identify objects under different lighting conditions. Contrast Adjustment: Adjusting the image's contrast helps the model learn to differentiate objects with varying contrast levels. By applying these transformations to the training images, the model can learn to be more invariant to these variations and improve its performance on unseen data. It is essential to carefully select and tune these augmentation techniques based on the specific characteristics of the dataset and the task at hand to achieve optimal results, as shown in Fig. 2.

The processing is split into three primary components. Initially, a GP filter is coupled with a feature extraction model to process the images efficiently. The GP filter was trained, and it returned an evolved expression. The best arithmetic expression was used for testing. After that, the developed and trained GP expression is applied to test images to obtain optimal results. Using this approach, the applied kernel center value is replaced by the favorable value across the whole image. In the end, the best projected evolved function is computed. Below, Equation (1) depicts the evolving faction.

$$F_n(X) = *((\log(0.731), \log(\sin(\log(f_3))))), /(\log(\cos(\log(f_1))), -\log(\sin(f_1)), \\ *(\text{power}(f_2, +(* (0.731, f_2), f_1)), /(\log(\cos(\log(f_3))))), *(\log(\log(f_3)), - \\ (\text{power}(f_1, /(+ (0.431, f_2)), /(\log(\sin(\log(f_1))))), +(\log(\sin(-(\sin(0.731), \\ \log(+ (\log(\cos(f_3)), /(\cos(\log(\sin(f_1))))), *(\sin(\log(f_3))), -(\text{power}(f_2, +(/ (0.314, f_3), f_1))), \\ *(\sin(\log(\log(f_1))), /(\cos(\sin(*(\log(0.731), \\ \log(+ (\log(\log(f_3))))), *(\sin(\log(\sin(f_2))), /(\log(\log(f_2)), -(\log((\sin(f_1))), /(\log(f_3), f_2))))))))))))), *(\cos(\log(\text{power}(f_2))), -(\log(\text{power}(f_3))), - \\ (\log(\cos(\log(f_1))), /(\log(\log(f_3), f_2))))))))), -(/(\sin(\log(*(\sin(0.731), \\ \log(+ (\cos(f_2), f_3))))), /(\sin(\log(\log(f_3))), \sin(f_1)))))) \\ (1)$$

limited variations of health image data, enhancing the DL model's resilience and mitigating overfitting. In image processing and computer vision, data augmentation is a technique used to artificially increase the

After processing the datasets, the holdout validation technique was used to carry out validation. Image training was done utilizing a variety

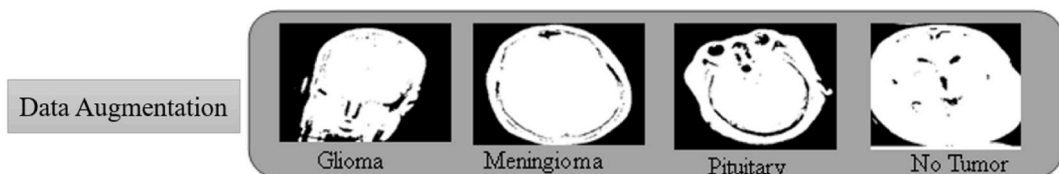


Fig. 2. Illustration of the data augmentation process applied to MRI scans, displaying examples of glioma, meningioma, pituitary tumors, and no tumor.

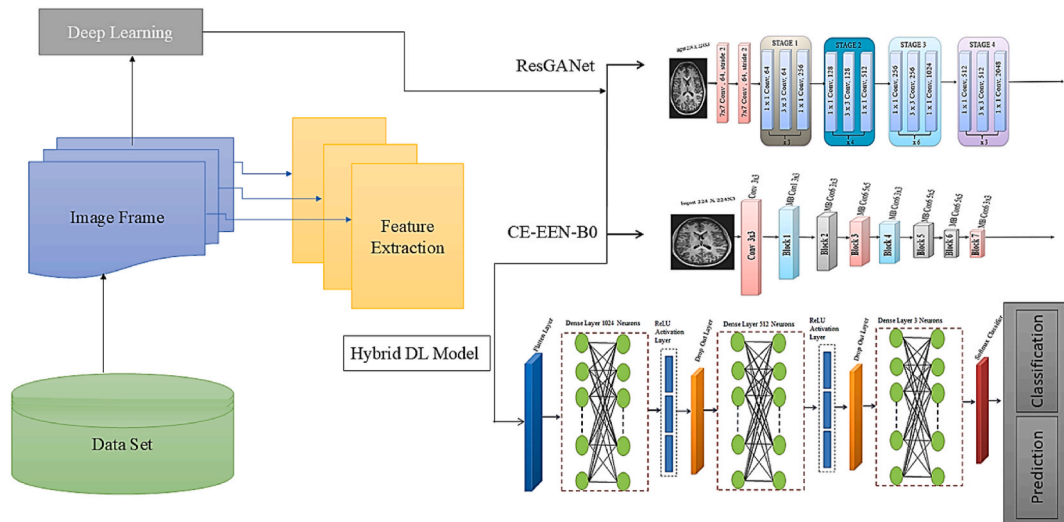


Fig. 3. Proposed hybrid DL model.

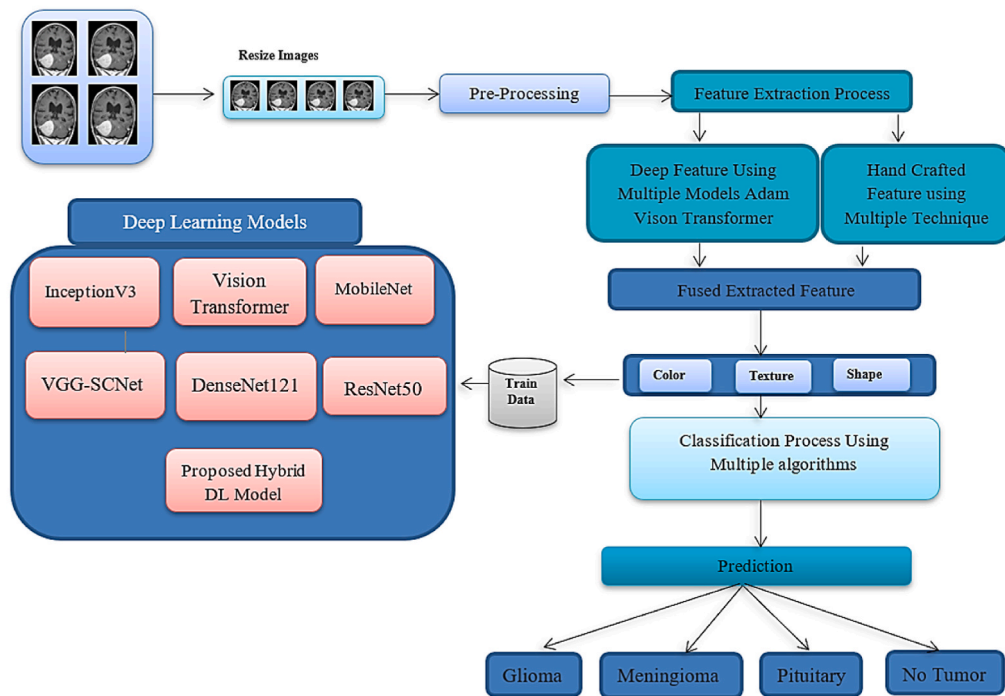


Fig. 4. Illustration of MRI Brain Tumor Classification using DL Models and Feature Extraction Techniques.

of DL algorithms. The dataset was divided into 80 % for training, 10 % for testing, and 10 % for validation. The objective of our study was to verify the authenticity of four specific categories of brain images: meningioma, glioma, pituitary, and no tumors. To verify the results, many metrics were studied, including accuracy, precision, recall, F1-score, error rate, specificity, Cohen's kappa, and loss. Fig. 3 shows the proposed approach, while Fig. 4 shows the step-by-step process of all DL models used in the methodology of this study.

3.2. Hybrid system model

3.2.1. Domain adaptation

Domain Adaptation is a technique in DL where a pre-trained model serves as the foundation for a new task. Rather than starting from scratch, one can leverage the features and weights of an existing model

to adapt it for a new task (Noor et al., 2020). The pre-trained model has often been trained on a large, labelled dataset for a related task, such as image classification. The learned features of this model provide an advantage when addressing a new related task with a smaller dataset (Horry et al., 2020). Domain adaptation proves particularly beneficial when there is limited data for the new task. This is because the pre-trained model has already acquired useful feature representations from its original training (Chaudhary et al., 2021). To avoid the problem of vanishing gradients, ResGANet is used as Model 1, which tackles the problem of vanishing gradients in deep networks using residual blocks. Moreover, to enhance efficiency and accuracy, the CE-EEN-B0 model family enhances efficiency and accuracy by harmonizing model depth, width, and resolution through a compound scaling strategy (Khan et al., 2024), and it is used as Model 2. The last layer of both models is removed and connected using the ensemble DL approach.

3.2.2. Hybrid deep learning technique

The hybrid DL model is a DL approach primarily used to extract features from two or more different models. In medical image classification, the Hybrid DL approach harnesses the strengths and diversity of each model. Moreover, it enhances the precision and robustness of the overall system model. Furthermore, the hybrid DL system model merges multiple DL models to improve overall performance for each class to detect critical conditions, and hybrid DL models can incorporate various models, each trained on different input data, such as MRI scans, and predict the best accuracy. Moreover, domain adaptation in the Hybrid DL model refines each model. All model's predictions are often aggregated by averaging to produce the final prediction, as shown in the following equation 2

$$\mathfrak{P} = \text{mod}[M_1(x), M_2(x)] \quad (2)$$

The determination of the class label P follows a majority voting scheme, wherein P is assigned the value of M1 (x) to the predictive deep hybrid CNN model if M1 (x) is greater than or equal to M2 (x), and M2 (x) is selected if M2 (x) surpasses M1 (x) as given in conditional equation 3

$$\mathfrak{P} = \begin{cases} M_1(x) & \text{if } M_1(x) \geq M_2(x) \\ M_2(x) & \text{if } M_2(x) > M_1(x) \end{cases} \quad (3)$$

using the unique strengths of models trained on different data, the deep hybrid CNN system model enhances accuracy and minimizes the occurrence of false positives and negatives. Nevertheless, the deep hybrid CNN system model should not be exclusively relied upon for diagnosis but complement other diagnostic tools. Hybrid methods amalgamate predictions from M₁ and M₂ models to identify the predominant class. In this research, CE-EEN-B0 and ResGANet classifiers were integrated using a deep hybrid CNN system model for the precision prediction of different diseases in the medical image data, as depicted in Fig. 3.

$$\mathfrak{L}_{WKL} = 1 - \frac{\sum_{i,j} w_{ij} \mathfrak{D}_{ij}}{\sum_{i,j} w_{ij} \mathfrak{E}_{ij}} \quad (4)$$

In equation (4), $\mathfrak{D}_{ij}$ shows the actual agreement, a matrix that shows the likelihood with which the raters of the models achieved a consensus on a particular class. $\mathfrak{E}_{ij}$ is the predicted recognition of a matrix that shows how likely it is for the raters of the models to agree based on uncertainty alone. While the variance behavior between labels i and j is given a weight of w_{ij} . Usually, the weight is based on how much difference there is between the two models. The proposed Hybrid DL algorithm is given below

Algorithm. The Proposed Hybrid DL Algorithm

- [1] Required: Annotated training samples $S_{j,j} = 1, \dots, n$, with labels $\psi_j \in \{\epsilon_1, \dots, \epsilon_c\}$
- [2] Ensure: Decision via Hybrid DL Model with advanced mathematical weighting and regularization
- [3] Initialization:
- [4] Assign initial weights: Set initial weights for samples $\omega_j^{(0)} = \frac{1}{n}, j = 1, \dots, n$
- [5] Initialize Diversity and Feature Quality Metrics
- [6] Define a diversity metric $D_m = 0$ for each classifier f_m
- [7] Initialize feature extraction quality metric D_{f_m} , which measures the average information gained from each convolutional feature map and sets $D_{f_m}^{(0)} = 1$
- [8] for $\omega = 1$ to T do
- [9] Train the classifier f_m :
- [10] Use a weighted training set with L2 regularization to penalize a complex model:
- [11] $L(f_m) = \sum_{j=1}^n \omega_j^{(m-1)} \mathcal{L}(f_m(S_j), \psi_j) + \frac{\beta}{2} \|\theta_m\|^2$
- [12] Where $\mathcal{L}(f_m(S_j), \psi_j)$ is the loss function, β is the regularization parameter, and θ_m are the weights of the model f_m
- [13] Evaluate the Weighted error e_m of classifier f_m :
- [14] $e_m = \frac{\sum_{j=1}^n \omega_j^{(m-1)} \cdot 1(f_m(S_j) \neq \psi_j)}{\sum_{j=1}^n \omega_j^{(m-1)}}$
- [15] Where $1(\cdot)$ is the indicator function

(continued on next column)

(continued)

- [16] Determine classifier weight α_m :
- [17] Adjust α_m based on the classifier error, regularization term, and diversity metric:
- [18] $\alpha_m = \log\left(\frac{1 - \epsilon_m}{\epsilon_m}\right) + \lambda + \gamma D_m$
- [19] Where λ is a penalty parameter, and γ controls the influence of the diversity metric
- [20] Update sample weights:
- [21] Adjust sample weights using an exponential update rule, penalizing incorrectly classified samples
- [22] $\omega_j^{(m)} = \omega_j^{(m-1)} \cdot \exp(\alpha_m \cdot 1(f_m(S_j) \neq \psi_j) + \delta \cdot \|f_m(S_j) - \psi_j\|^2)$
- [23] Where δ Controls the additional weights penalty based on the classification margin
- [24] Enhance Hybrid Diversity:
- [25] Compute D_m based on the cosine similarity between the classifier f_m and the hybrid model F_{m-1} :
- [26] $D_m = 1 - \frac{\langle f_m, F_{m-1} \rangle}{\|f_m\| \cdot \|F_{m-1}\|}$
- [27] This metric ensures each classifier contributes unique information to the hybrid.
- [28] Normalize Weight:
- [29] Normalize weights to prevent their exponential growth:
- [30] $\omega_j^{(m)} = \frac{\omega_j^{(m)}}{\sum_{k=1}^n \omega_k^{(m)}}$
- [31] end for
- [32] Hybrid Model Decision:
- [33] Non-Linear Aggregation of Classifier Outputs:
- [34] For each class ϵ_r , compute a weighted confidence score using a softmax function:
- [35] $U_i = \sum_{m=1}^T \gamma_m \cdot Q_{f_m} \exp(\alpha_m \cdot \sigma(f_m(\hat{S}) = \epsilon_r))$
- [36] Where $(\cdot)$ is the sigmoid function to map outputs to a confidence level, γ_m is a scaling factor and Q_{f_m} is the feature extraction quality metric for f_m , updated as
- [37] $Q_{f_m} = \frac{1}{d} \sum_{k=1}^d H_k(f_m(S_j))$
- [38] where $H_k(\cdot)$ is the entropy of each feature map in layer k, and d is the number of feature maps
- [39] Dynamic Weighting Based on Feature Extraction Quality:
- [40] Refine γ_m for each classifier, weighted by the gradient norm of feature maps:
- [41] $\gamma_m = \frac{1}{\|\nabla F_m\|} \sum_{j=1}^n \omega_j^{(m)} \cdot Q_{f_m}$
- [42] Assign the test sample $\hat{S}$ to the class ϵ_r with:
- [43] $\epsilon_r = \text{argmax}_i U_i$

The described algorithm is a hybrid DL method that integrates different classifiers to make decisions regarding data classification. At the beginning, every training sample is assigned the same weight. The approach sequentially trains a classifier on the weighted training set and assesses its errors. The error is used to determine a weight for the classifier, which is then used to apply a complexity penalty to prevent overfitting. The weights of the samples are subsequently adjusted, assigning greater importance to the misclassified samples. This ensures that the subsequent classifier prioritizes these samples. The sample weights are normalized, and then multiple classifiers are integrated using a nonlinear algorithm to calculate a dynamic score for each potential class label. The ultimate determination for a test sample is the category with the highest score, representing the ensemble's combined decision, considering each classifier's dependability and difficulty in accurately categorizing each sample.

3.3. Grad-CAM technique

Grad-CAM is the abbreviation for Gradient-weighted Class Activation Mapping, and it is a substantial visualization method that has transformed the field of brain tumor identification. Grad-CAM allows doctors and researchers to acquire vital insights into the decision-making process of DL models by visually representing the brain areas that are most important for a specific class prediction. Grad-CAM detects brain tumors by presumably indicating the exact locations of the brain image that most represent a tumor, enabling more accurate diagnosis and better treatment outcomes. Furthermore, Grad-CAM can correct misdiagnoses, false positives, or false negatives for better diagnostic accuracy and patient prognosis.

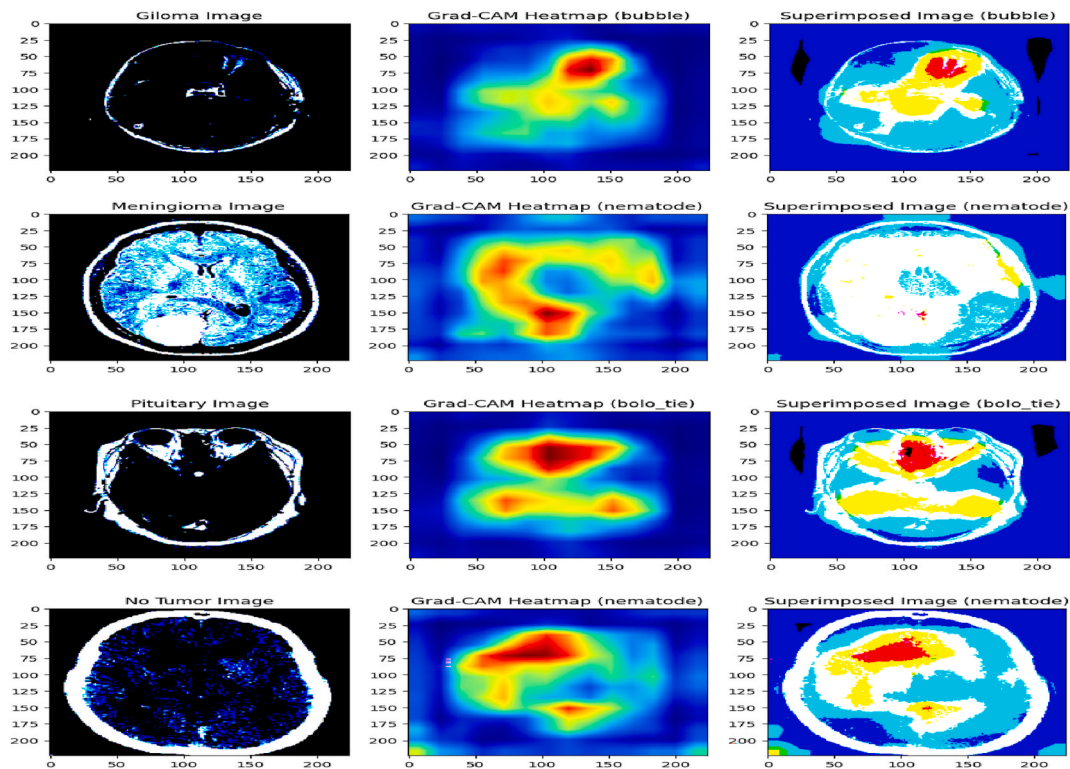


Fig. 5. Grad-Cam Technique for visualizing our proposed system model.

Grad-CAM, a visualization technique, is applied to test the brain MRI images. After the results, it is clearly shown in Fig. 5 that the proposed system model more actively focuses on the target part of the images, such as the tumor region.

3.4. Regression analysis extension

In addition to classification, the hybrid model was extended to perform a regression task aimed at estimating a continuous confidence score related to tumour severity. After the final dense layers of the classification path, a parallel output with a linear activation function was added to predict continuous values. The model was trained using Mean Squared Error (MSE) loss and evaluated using MSE, Mean Absolute Error (MAE), and R^2 score. This allows the model to not only identify tumor type but also predict severity on a scale from 0 to 1, offering enhanced diagnostic insight (Renugadevi et al., 2023).

3.5. Data analysis and statistical evaluation

Statistical analysis was performed using SPSS software (Version 24). Descriptive statistics, including means and standard deviations, were computed for all variables. An independent samples *t*-test was conducted to compare the means between the two groups. The normality of the data was assessed using the Shapiro-Wilk test, with a *p*-value of less than 0.05 considered statistically significant. Pearson correlation analysis was performed to assess the relationships between different variables.

4. Experimental results

The Hybrid system model underwent rigorous testing for many months, utilizing projects conducted by our research department. The main objective was to confirm the initial goal, which was to develop a user-friendly platform that does not require advanced expertise and to validate its compatibility with various devices, ensuring that an average user can effectively utilize the entire solution without facing technical

difficulties or performance degradation. Using the MRI images dataset, we evaluated the performance of different DL methods in recognizing four distinct classes, including No tumor Conditions. We used InceptionV3, Vision Transformer, MobileNet, VGG-SCNet, DenseNet121, and ResNet50 for validation and comparison of the proposed Hybrid-based DL model integrating two advanced CNN architectures, CE-EEN-B0 and ResGANet.

4.1. Environment setup

The proposed method used keras-supported TensorFlow in the Python 3.7.1 framework. A personal computer was equipped with an Nvidia GeForce RTX-3070 GPU, 32 GB of RAM, Microsoft’s Windows 10 operating system, and the CUDA toolkit version 11.0 with cuDNN version 8.0 to conduct experiments. The experimental findings showcased the efficacy of the suggested system model approach, exhibiting a substantial enhancement compared to the current cutting-edge technologies.

4.2. Dataset collection

Our dataset was sourced from publicly available data on Kaggle.com, especially for the diagnosis of brain tumors. We constructed the dataset using MRI images. We opted for MRI images for our study because they are the most effective method for identifying brain tumors. Our dataset comprised four distinct brain tumor categories: Glioma (1605 images),

Table 2
Detailed description of the brain tumor dataset.

Classes	Number of Images	Training Set	Testing Set
Glioma	1605	1284	321
Meningioma	1605	1284	321
Pituitary	1605	1284	321
No tumor	1605	1284	321
Total Images	6420	5136	1284

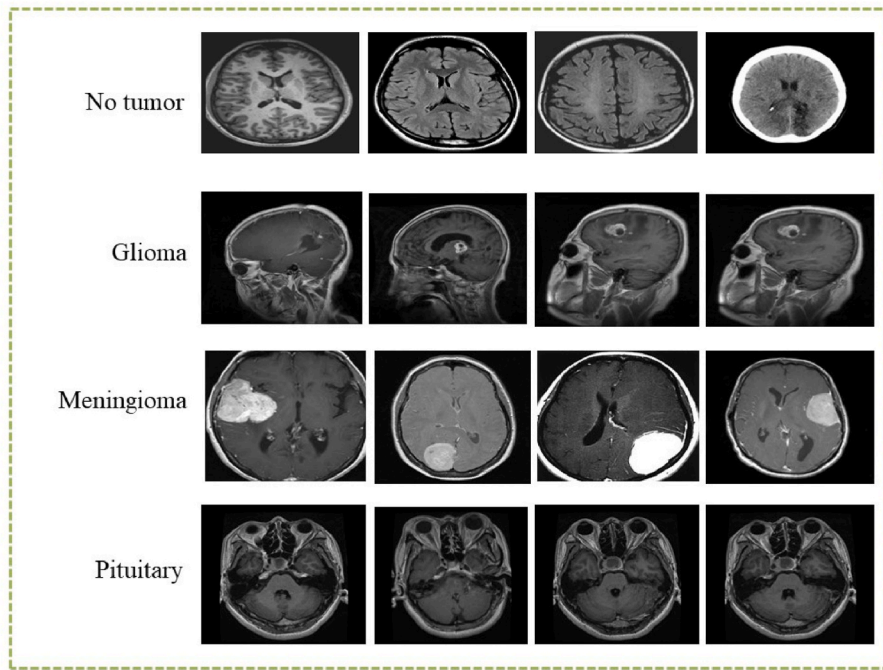


Fig. 6. Mri images of brain tumors.

Meningioma (1605 images), Pituitary (1605 images), and No tumor (1605 images). Altogether, we integrated 6420 MRI images into our dataset. The dataset distribution is presented in Table 2 in (Woźniak et al., 2023), while Fig. 6 in (Woźniak et al., 2023) shows the MRI images categorized by the different brain tumor types.

4.3. Pre-processing of the dataset

Pre-processing is a crucial step in which the data undergoes adjustments to become suitable for training. Because the MRI images were

acquired from a patient database, many exhibited inadequate clarity and quality. At this phase, we normalized these images for better subsequent processing.

4.4. Data division and enhancement

Despite the limited nature of our dataset, which solely consisted of MR images, it is widely recognized that deep neural networks need substantial amounts of data to yield significant results. The dataset contained a sum of 6420 MR images. We allocated 80 % for training

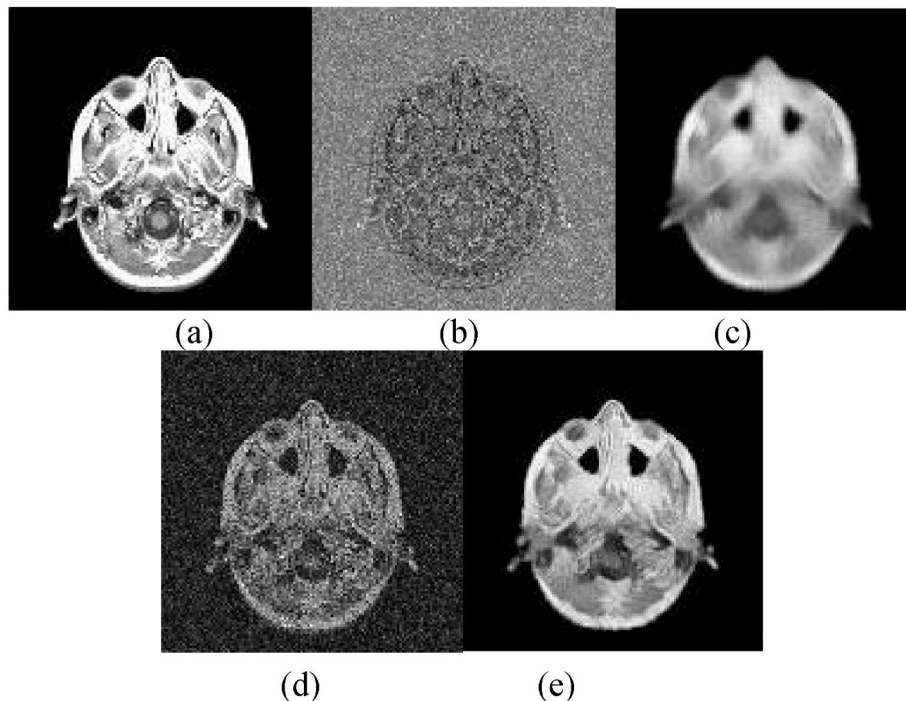


Fig. 7. (a) Original image (b) Rician noise affected image (c) De-noised with Fusion method (d) De-noised with NLM (e) De-noised with proposed GP.

while dividing the remaining 20 % equally for testing and validation. Data augmentation, a method to increase the dataset's size, can enhance training. By using this, we amplify the model's learning potential. Specifically, we augmented data by mirroring the MR images and implementing modifications such as rotation, width and height adjustment, and magnification. We then validated the datasets using a holdout validation method.

4.5. Validation technique

Our 6420 MRI image dataset proved pivotal in selecting an optimal validation strategy. We utilized the holdout validation technique, allocating 80 % of the data for training and reserving 20 % for testing. This prevalent method typically delivers effective outcomes. The essence of the holdout approach is to divide the dataset into training and testing segments, accelerating the model's training pace. While the training subset educates the DL model, the testing subset gauges its efficacy. In the holdout approach, a random 80 % of the dataset constitutes the training segment, and the remaining 20 % serves as the testing segment. The model acquires knowledge from the training set and undergoes assessment on the testing set to evaluate its capabilities. An 80 % split for training provides the model with a broader learning foundation, improving its capacity to generalize to unexpected data. Nevertheless, the testing subset might not wholly mirror the complete dataset, potentially skewing the performance assessment.

4.6. Evaluation metrics

We concentrated on the real-time assessment of model performance in supervised classification tasks in medical imaging, emphasizing the specific metrics employed.

Figs. 7 and 8 show the comparative results of our propose model with SOTA methods. As seen in Figs. 9 and 10, it is essential to note that the model's testing data has not been previously encountered while analyzing the efficiency of a trained model through accuracy. The analysis procedure requires many critical measures to comprehensively analyze the model's efficacy. These measures include F1-score, precision, error rate, overall accuracy, sensitivity (or recall), specificity and Cohen's Kappa. The confusion matrices are presented in Fig. 11 is utilized to get deeper insights into the model's predictive capacities for upcoming occurrences. It comprises critical components like true positive (TP), true negative (TN), false positive (FP), and false negative (FN), which may be utilized successfully to find out the comprehensive set of performance assessments for real-time system models.

TP and TN indicate that the model performs correct prediction for positive and negative results significantly. Nevertheless, FP events happen when the model incorrectly predicts the positive class, specifying the areas where its positive class predictions can be improved most. Likewise, FN cases are indicative of the model's misclassification of the negative class, and, thus, the model may have errors in the prediction of the negative class, where it actually could be correct, which

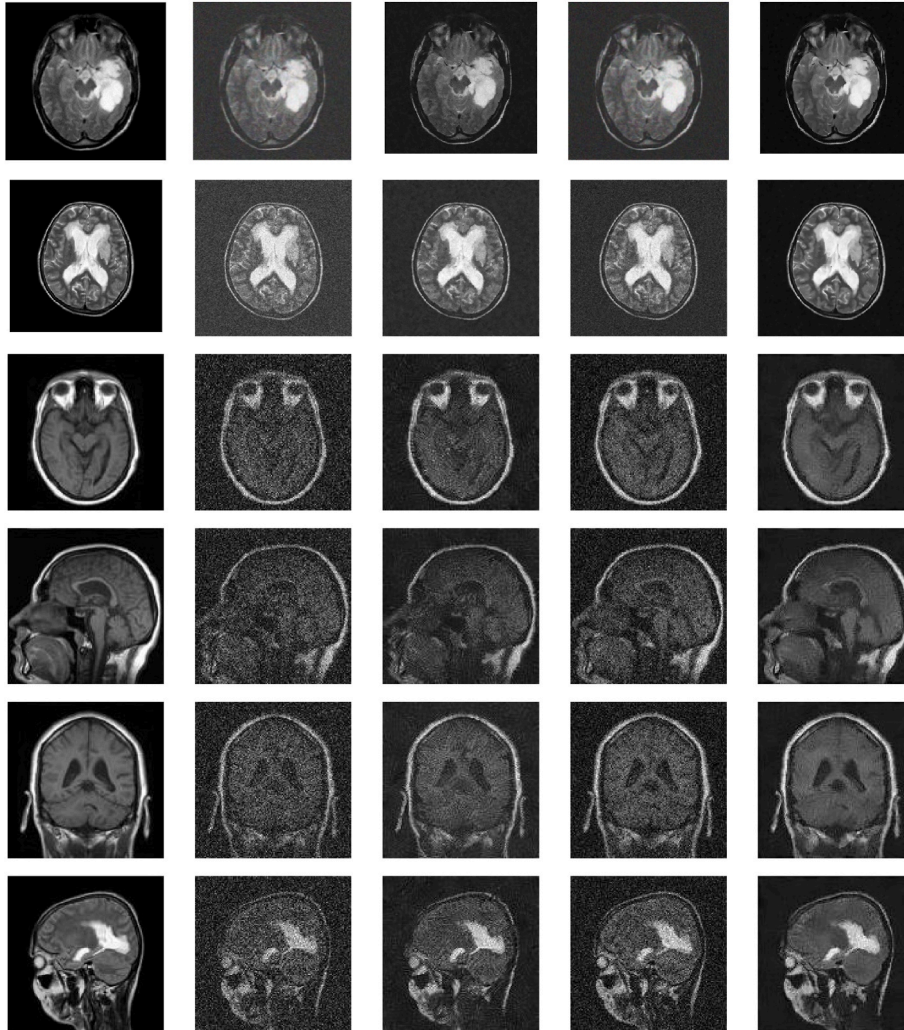


Fig. 8. (Col-1) original images, (Col-2): Rician noise affected images, (Col-3): De-noised with NLM, (Col-4): De-noised with Fusion method, (Col-5): De-noised with proposed GP.

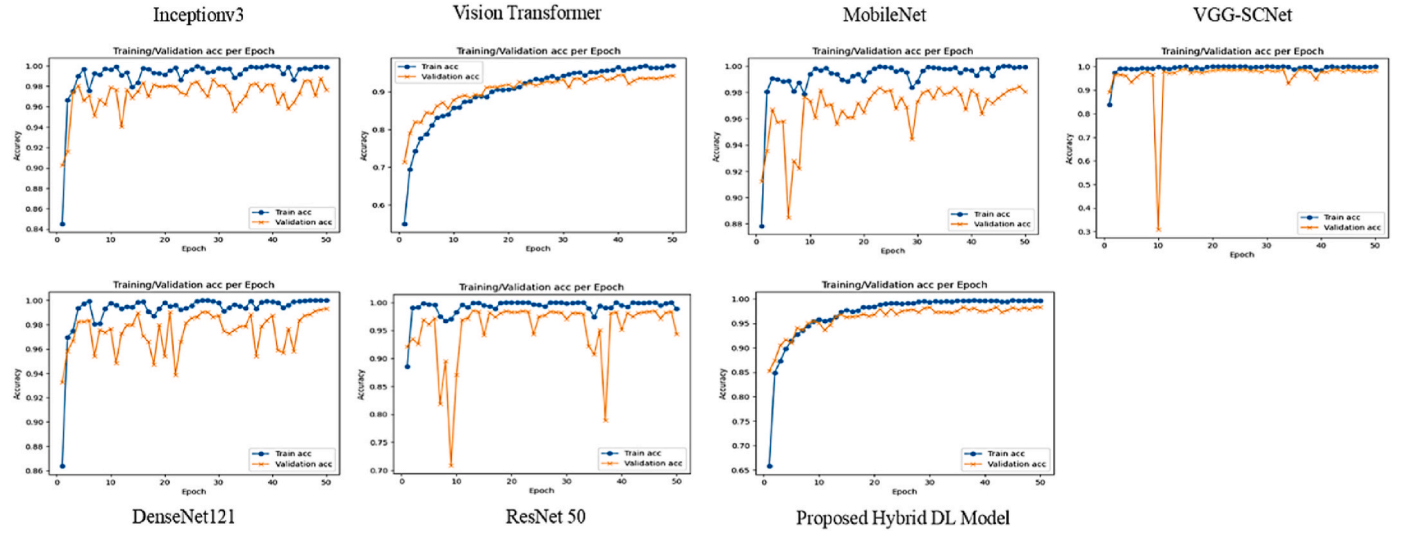


Fig. 9. Accuracy vs Epoch Graph for InceptionV3, Vision Transformer, MobileNet, VGG-SCNet, DenseNet121, ResNet50 and Proposed DL Model.

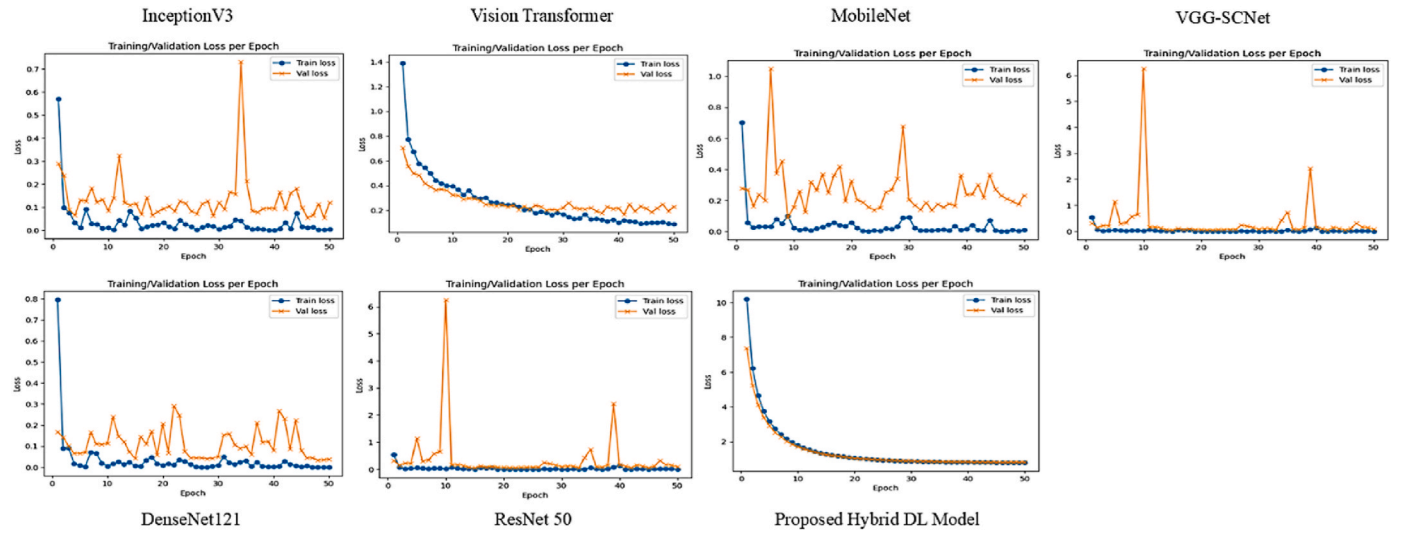


Fig. 10. Loss vs Epoch Graph for InceptionV3, Vision Transformer, MobileNet, VGG-SCNet, DenseNet121, ResNet50 and Proposed DL Model.

gives us some ideas of areas that may need to be improved to enhance its whole predictive accuracy. The comprehensive evaluation method assures a thorough understanding of the model's performance across several aspects, facilitating informed decision-making and potential preemptive modifications to the model.

Equations (5) and (6) are used to systematically assess a particular test dataset's accuracy and error rate. This process is thoroughly discussed in the relevant sections. Moreover, describing the evaluation process flow and having equations that specify the precision, accuracy, recall, and F1 score is necessary. This way, the performance analysis will include more aspects and be more detailed. This method, which can be adjusted, gives a deep understanding of the model's performance, which allows both describing and exploring its prediction capabilities in detail.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (5)$$

$$\text{Error Rate} = \frac{FP + FN}{\text{Total}} \quad (6)$$

The word sensitivity, or recall, in equation (7) is often called the TP rate. In addition, equation (8) depicts the relationship between the TN and FN values that help to determine the model's specificity.

$$\text{Sensitivity or Recall} = \frac{TP}{TP + FN} \quad (7)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (8)$$

In addition, equation (9) calculates the ratio of expected TP results to the total number of actual positive outcomes to compute the model's accuracy. The prevalence refers to the frequency at which the true event occurs in our trained model, as specified in equation (10).

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

$$\text{Prevalance} = \frac{\text{Actual Yes}}{\text{Total}} \quad (10)$$

Similarly, equation (11) determines the null error rate, which serves as a metric and benchmark for assessing the model's error level while consistently predicting the majority class.

$$\text{Null Error Rate} = \frac{\text{Actual No}}{\text{Total}} \quad (11)$$

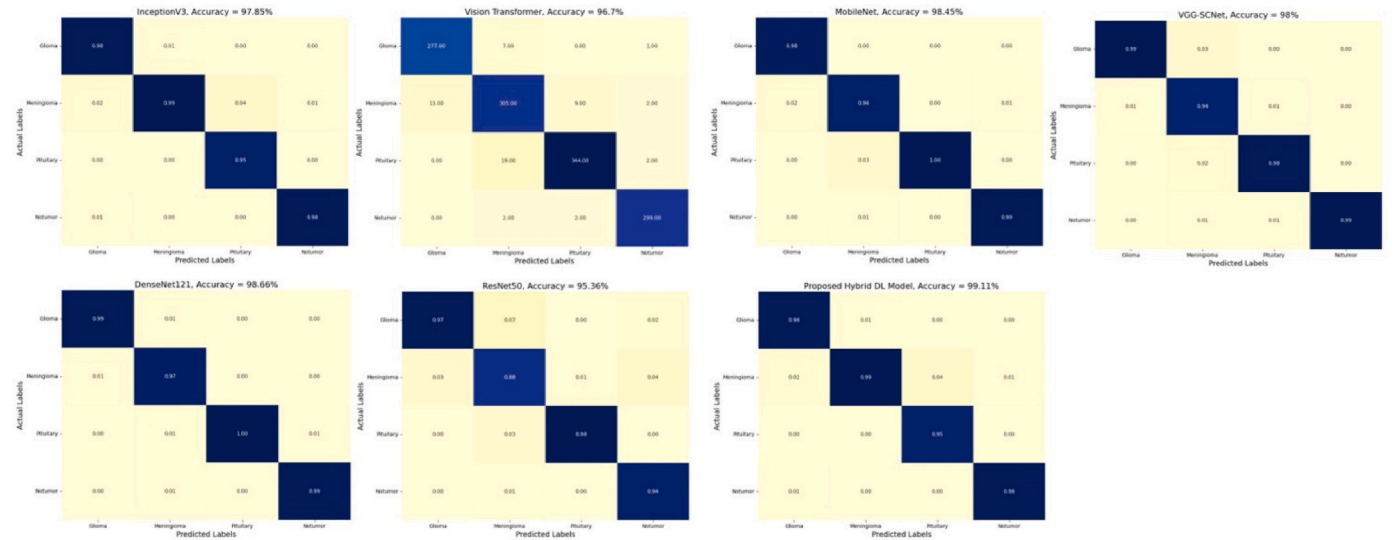


Fig. 11. Tested confusion matrix of all architectures with brain tumor MRI dataset.

Kappa (K) or Cohen's Kappa is a metric that determines the degree of agreement between an evaluation and its predicted performance, accounting for randomness. A model exhibits a high Kappa score when there is a substantial disparity between accuracy and the null error rate. The definition of K is represented by equation (12):

$$\text{Cohen's Kappa} = \frac{P_o - P_e}{1 - P_e} \quad (12)$$

Where P_o stands for observed agreement, which is calculated as the proportion of diagonal elements to the total number of instances, while P_e is the expected agreement and is formulated in equation (13);

$$P_e = \sum_{i=1}^k (P_{t_i}) \cdot (P_{p_i}) \quad (13)$$

Where P_t stands for the true labels that are expressed by adding the probabilities of every row in the confusion matrix and then dividing them by the total number of occurrences, whereas P_p refers to the predicted labels, which are obtained by adding the probabilities for each column of the confusion matrix and dividing them by the total number of occurrences. k is the number of classes in the above equation.

Additionally, the F1 score is a mathematical calculation of precision and recall, as shown in equation (14), and is usually called the F score or the F measure. F1 may be more suitable than accuracy when dealing with uneven class distribution. The maximum accuracy is obtained

Table 3
Performance evaluation on brain tumor images dataset.

Classifiers	Evaluation Parameters	Classes				Overall Accuracy	Error Rate	Cohen's Kappa	Average F1Score
		Glioma	Meningioma	Pituitary	No Tumor				
InceptionV3	Precision	0.94	0.99	0.98	0.99	0.9785	0.028	0.9589	0.9794
	Recall	0.96	0.98	0.97	0.99				
	F1-Score	0.99	0.95	0.98	0.97				
	Specificity	0.98	0.99	0.99	1.00				
Vision Transformer	Precision	0.99	0.93	0.95	0.96	0.967	0.056	0.9799	0.965
	Recall	0.98	0.91	0.95	0.99				
	F1-Score	0.96	0.97	0.99	0.94				
	Specificity	0.98	0.99	0.99	0.97				
MobileNet	Precision	0.98	1.00	1.00	0.98	0.9845	0.019	0.9810	0.98
	Recall	0.99	1.00	0.99	0.96				
	F1-Score	0.96	0.99	0.99	0.98				
	Specificity	0.99	1.00	0.98	1.00				
VGG-SCNet	Precision	0.98	0.96	0.99	0.98	0.98	0.0213	0.9635	0.9725
	Recall	0.95	0.99	0.99	0.98				
	F1-Score	0.98	0.99	0.97	0.95				
	Specificity	0.99	0.98	0.98	1.00				
DenseNet121	Precision	0.99	0.98	0.99	0.99	0.9866	0.0108	0.9805	0.985
	Recall	0.97	0.99	0.99	1.00				
	F1-Score	0.98	0.99	0.98	0.99				
	Specificity	0.99	0.99	1.00	0.98				
ResNet50	Precision	0.93	0.90	0.97	0.97	0.9536	0.068	0.9361	0.95
	Recall	0.96	0.89	0.95	0.97				
	F1-Score	0.91	0.96	0.98	0.95				
	Specificity	0.95	0.98	0.99	0.98				
Proposed Hybrid DL Model (CE-EEN-B0-ResGANet)	Precision	0.99	1.00	1.00	0.99	0.9911	0.01	0.9910	0.996
	Recall	1.00	0.99	0.99	1.00				
	F1-Score	0.99	1.00	0.99	1.00				
	Specificity	1.00	0.99	0.99	1.00				

when FN and FP costs are equal. Precision and recall should be considered when there is a big difference in the costs between FP and FN.

$$F - \text{measue or } F1 - \text{score} = \frac{2TP}{2TP + FP + FN} = \frac{2.(Prec.Recall)}{Prec + Recall} \quad (14)$$

Moreover, the method of four classification is used in our study, so it is of paramount importance to compute the average F1 score for the total N classes, which can be found by formula 15 given below:

$$F1_{macro} = \frac{1}{N} \sum_i F1_i \quad (15)$$

Figs. 7 and 8 demonstrate that the proposed GP method achieves superior performance compared to existing SOTA approaches.

5. Discussion

An analysis was conducted on the brain tumor MRI dataset to compare the performance of different advanced DL models, including InceptionV3, Vision Transformer, MobileNet, VGG-SCNet, DenseNet121, ResNet50, and the proposed Hybrid DL Model classification algorithms. The results of this analysis are presented in Table 3. Table 3 displays the performance of these models in terms of Accuracy, Specificity, Precision, Recall, F1-Score, Error Rate, Cohen's Kappa, and Average F1 Score. After analyzing the methods of InceptionV3, Vision Transformer, MobileNet, VGG-SCNet, DenseNet121, ResNet50, and the Hybrid DL Model, it was observed that the proposed Hybrid DL approach outperformed the other DL models, as shown in Table 3. The proposed model achieved a validation accuracy of 99.11 %, an error rate of 0.01, a Cohen's Kappa of 0.9910, and Precision, recall of 99.5 %, an average F1 Score of 99.6 % with a Specificity of 99.75 %. Regarding all CNN architectures, the proposed system model achieved the most optimal classification and testing times, which were less than 100 ms. By combining ResGANet with CE-EEN-B0, the research leverages the key features of both models. This integration enhances feature extraction and classification, resulting in superior performance. The synergy of ResGANet's deep architecture and CE-EEN-B0's efficient scaling expands the model's capacity to handle complex data, offering improved precision and robustness compared to each individual model.

Beyond classification performance, we extended our hybrid model to include a regression output path to predict continuous tumour severity scores based on the extracted features. This addition enhances the clinical value of the system by allowing it to estimate a quantitative severity level, which can support prioritization and risk-based treatment

decisions. The regression model achieved strong results, with an R^2 score of 0.942, a low MSE of 0.018, and a MAE of 0.09, demonstrating a high degree of alignment between predicted and actual values. These results indicate that the proposed model is capable of learning not only categorical tumor types but also subtle variations in tumor characteristics on a continuous scale.

In Figs. 9 and 10, the validation accuracy and validation loss graphs are presented in comparison to the training accuracy and training loss, respectively, for the InceptionV3, Vision Transformer, MobileNet, VGG-SCNet, DenseNet121, ResNet50, and proposed Hybrid DL models. Blue lines represent the accuracy of the training, while orange lines represent the accuracy of the validation. The proposed system model achieved the highest validation accuracy at 99.11 %. DenseNet121 followed closely with a validation accuracy of 98.66 %, MobileNet with 98.45 %, VGG-SCNet with 98 %, InceptionV3 with 97.85 %, Vision Transformer with 96.7 %, and ResNet50 with the lowest at 95.36 %. 50 epochs were utilized to implement these models, with a batch size of 32. The Adam optimizer was employed as well. Based on the accuracy graph study, the proposed system model outperformed the other models. This was owing to its validation accuracy curve being superior to the training accuracy curve, with no problems of overfitting or underfitting.

The classification performance of all DL models using the confusion matrix for the test data is shown in Fig. 11. We have predicted four possible classes: glioma, meningioma, pituitary, and no tumor. The classifier was tested on 1284 images, with 321 for each class.

The evaluation included preprocessed denoised images from the brain MRI dataset, shown in Fig. 12. It displays the confidence scores for brain images featuring glioma, meningioma, pituitary, and no tumor. These scores are utilized for experimental purposes. Fig. 12 represents a different confidence score used to verify the accuracy of the actual data and evaluate the proposed system model's performance.

6. Conclusion

Timely and accurate identification of brain tumors is critical in reducing overall mortality rates and improving patient outcomes. The diagnosis of brain tumors can be incredibly challenging due to the variances in tumor appearance, size and structure. Even with these challenges, MRI imaging is still one of the best ways available for early detection and classification of brain tumors. This study presents a novel Hybrid DL Model that merges the CE-EEN-B0 and ResGANet architecture, developed to accurately detect and classify brain tumors at an early stage. The suggested model demonstrated 99.11 % accuracy, outperforming existing SOTA models and successfully classifying glioma,

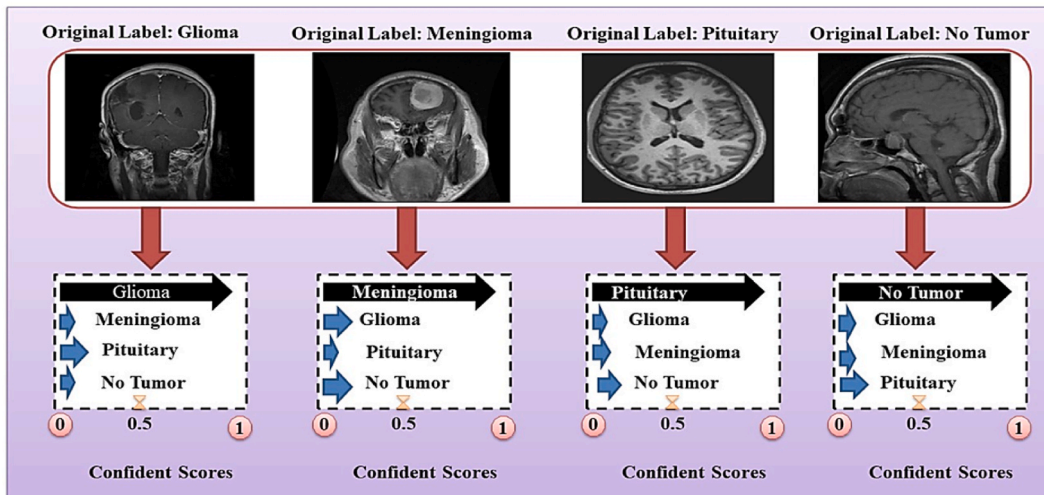


Fig. 12. The predicted results of evaluating brain MRI images utilizing the proposed system model.

meningioma, pituitary tumors, and no tumor.

In addition to classification, we extended the model with a regression output path to estimate continuous tumor severity scores. This regression extension provides valuable quantitative insights and presents a confidence score related to the severity of the tumor, which can assist in more informed clinical decision-making. The model performed strongly across key evaluation metrics, which include MSE, MAE, and R^2 , further confirming its potential for continuous risk assessment in clinical setting.

Using Grad-CAM to interpret the model's decision processes visually improves interpretability, thus increasing the reliability of its prediction for healthcare professionals. The combination of these innovations in classification and regression, along with the use of interpretable visualizations, positions this Hybrid DL model as a promising tool for automated, accurate brain tumor detection and diagnosis.

The results of this study demonstrate the capability of models to not only classify tumors with high accuracy but also provide useful insights into severity, which can improve treatment planning and patient care. The high accuracy and early detection skills of this model are expected to lead to more cost-effective treatment strategies by facilitating earlier interventions and potentially reducing the need for more intensive and expensive treatments at later stages. This can update healthcare policy by optimizing resources allocation for brain tumor diagnosis and management. Future work will emphasis on expanding the dataset used, filtering the model's scalability, and exploring its application to other medical imaging domains further to advance the accuracy and reliability of automated diagnostic systems.

CRediT authorship contribution statement

Rahim Khan: Writing – original draft, Resources, Funding acquisition, Formal analysis. **Sher Taj:** Software, Investigation, Conceptualization. **Zahid Ullah Khan:** Writing – review & editing, Methodology, Data curation. **Sajid Ullah Khan:** Supervision, Project administration, Data curation. **Javed Khan:** Writing – review & editing, Visualization, Formal analysis. **Tahir Arshad:** Writing – original draft, Validation, Methodology. **Sarra Ayouni:** Validation, Software, Data curation.

Availability of data and materials

The datasets used and/or analysed during the current study are available from the corresponding author upon reasonable request.

Funding

Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R896), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R896), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

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